

A New Research Journey

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Abstract

Write a concise summary of your research question, approach, main findings, and their implications here.

1 Introduction

Introduce the problem, summarize related work, and explain your contribution. Cite sources using BibTeX, for example [1].

2 How many dimensions does assistance need?

2.1 Motivation

Lower-limb exoskeleton controllers are dominated by two families. *Gait-phase* controllers detect gait events, estimate a scalar phase $\varphi \in [0, 1)$, and index a torque profile by it; they implicitly assume that locomotion is periodic and that assistance needs to know nothing beyond *where in the cycle* the limb is. *End-to-end* controllers map a window of sensor history to torque with a neural network and assume nothing.

Comparing the two directly is close to a tautology: the end-to-end controller observes strictly more than a scalar phase, so in-distribution it cannot lose. We therefore replace the comparison with a measurement:

How many free variables does the assistive target actually depend on?

2.2 A latent gait coordinate on $S^1 \times \mathbb{R}^k$

Let $w_t \in \mathbb{R}^{42}$ be the deployable observation: one leg’s hip flexion angle and angular velocity over the preceding 400 ms, sampled at 50 Hz. We compress it to a small code and predict the assistive target *from the code alone*:

$$(u_t, v_t, z_t) = \text{Enc}_\theta(w_t), \quad \varphi_t = \text{atan2}(v_t, u_t) \in S^1, \quad z_t \in \mathbb{R}^k, \quad (1)$$

$$\hat{y}_t = \text{Head}_\theta(\cos \varphi_t, \sin \varphi_t, z_t), \quad \hat{w}_t = \text{Dec}_\theta(\cos \varphi_t, \sin \varphi_t, z_t). \quad (2)$$

The target y_t is the biological (muscle-generated) knee moment $M_{\text{bio}}^{\text{knee}}$, i.e. the quantity a moment-matching controller must reproduce, low-pass filtered by the same exponential moving average the assist reward uses.

The circular factor S^1 encodes the periodicity of gait: one gait cycle is one revolution, and φ carries a single degree of freedom. The Euclidean factor $\mathbb{R}^k$ carries *everything that is not phase* — speed, turning, terrain, whether a transition is under way. Its width k is the experimental variable:

$$\underbrace{k=0}_{\text{assistance}=f(\varphi) \equiv \text{gait-phase control}} \quad \longrightarrow \quad \underbrace{k \rightarrow \text{large}}_{\text{unconstrained end-to-end}} \quad (3)$$

Sweeping k and recording where accuracy saturates measures the dimensionality of assistance directly; the two established families become the two ends of one curve rather than rivals.

2.3 Keeping the circle honest

Three constraints prevent the latent from degenerating. (i) The phase must *advance*: we penalise $\mathbb{E}[\text{relu}(-\Delta\varphi_t)]$ with $\Delta\varphi_t = \text{atan2}(\sin(\varphi_t - \varphi_{t-1}), \cos(\varphi_t - \varphi_{t-1}))$, otherwise φ may oscillate instead of circulating. (ii) The width of z is a hard budget — it is the swept variable — so phase information cannot be smuggled into z . (iii) After training we *ablate* φ by replacing it with a random angle; if accuracy does not collapse, the network is ignoring the circle and the experiment is void. The total loss is

$$\mathcal{L} = \underbrace{\mathbb{E}[(\hat{y}_t - y_t)^2]}_{\text{assist target}} + \lambda_{\text{rec}} \underbrace{\mathbb{E}[\|\hat{w}_t - w_t\|^2]}_{\text{structure}} + \lambda_{\text{mono}} \mathbb{E}[\text{relu}(-\Delta\varphi_t)], \quad (4)$$

with $\lambda_{\text{rec}} = 0.3$ and $\lambda_{\text{mono}} = 0.1$.

2.4 Protocol

All models are scored with *leave-clips-out* cross-validation: a motion clip never appears in both the training and the test split, folds are stratified by motion subcategory, and both legs of a clip share a fold. This is essential — in-distribution, a model with more inputs cannot lose, so only out-of-distribution generalisation carries information. We report

$$R^2 = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \bar{y})^2}, \quad (5)$$

for which $R^2 < 0$ means the predictor is worse than the training mean.

Three baselines share the protocol: a **heuristic-phase** controller (phase from the rising edge of the vertical foot contact force with a swing gate, then a cycle-average torque profile); a **plain autoencoder** of the same latent width but without the S^1 factor, which isolates the contribution of the topology; and an **unconstrained end-to-end** MLP on the same window, which serves as an upper reference.

2.5 Data

Three per-category expert human policies (walking, running, squatting) are rolled out without an exoskeleton over their training clips, recording bilateral hip and knee kinematics, muscle joint moments and foot contact forces. Actions are *sampled* from the policy and clipped to $[-1, 1]$; playing the distribution mean instead is not a benign simplification, because the stochasticity of these muscle policies is functional — deterministic playback over-activates and falls (completion 21/30, 18/20 and 0/10 for walking, running and squatting, versus 30/30, 20/20 and 7/10 when sampling). The final set is 32,958 samples grouped into twelve subcategories.

2.6 Results

Three observations follow (Table 1, Figs. 2–3).

The gap is in the parameterisation, not the dimensionality. At equal information — a single scalar — the heuristic phase reaches $R^2 = 0.096$ and the learned circular coordinate 0.578, a six-fold difference. The limitation of gait-phase control is therefore not that one dimension is too few, but that the hand-designed one dimension is a poor coordinate.

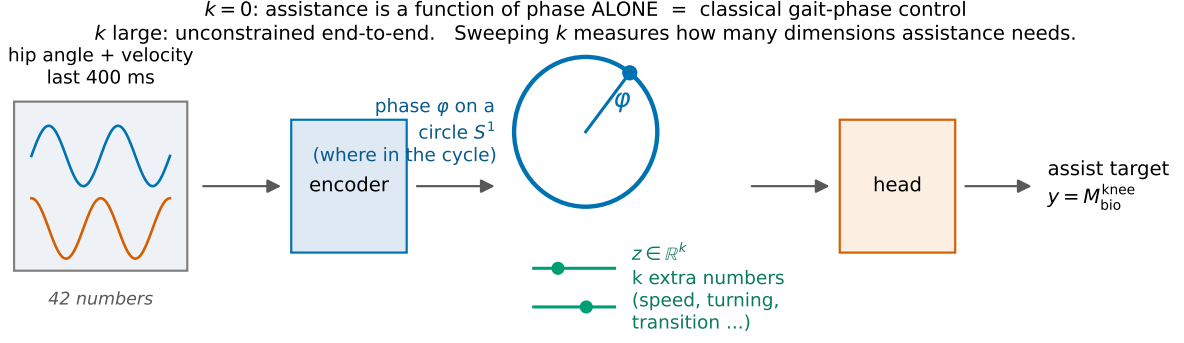


Figure 1: The observation window is compressed to a phase ϕ on a circle plus k free numbers; the assistive target is predicted from that code alone. $k = 0$ is gait-phase control; large k approaches unconstrained end-to-end control.

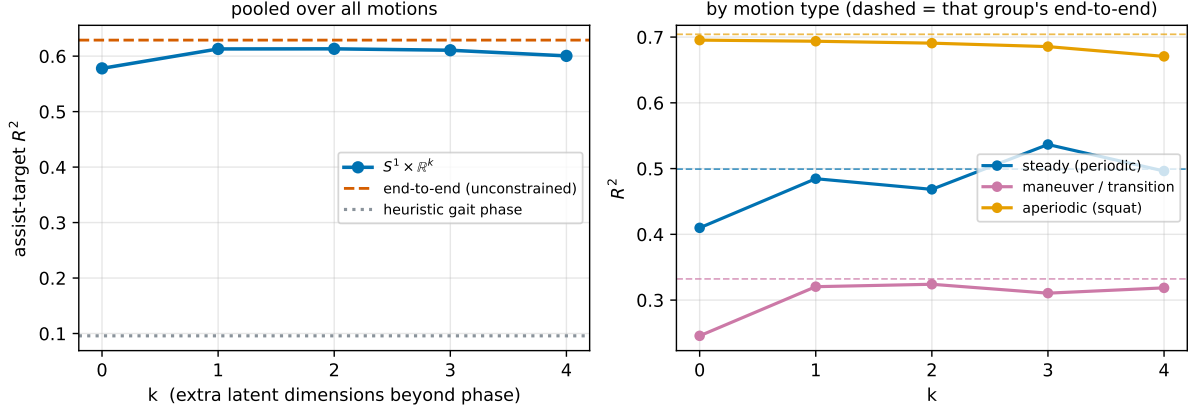


Figure 2: Assist-target R^2 against the number of non-phase latent dimensions k , pooled (left) and by motion type (right). Dashed lines are the unconstrained end-to-end reference, dotted the heuristic gait phase.

The heuristic phase fails outright away from steady gait. It is negative in eight of twelve subcategories and is usable only for steady walking (0.267). For squatting the phase is undefined on 97% of samples, which is the honest failure mode of a controller that must find a cycle before it can act.

Dimensionality requirements are task-dependent and hidden by pooling. Steady walking, steady running and squatting are near-saturated at $k = 0$, whereas start-stop, turning and circling reach only 47–59% of their end-to-end reference at $k = 0$ and recover most of the remainder at $k = 1$. No subcategory benefits beyond $k = 2$, where accuracy plateaus and then decays slightly. Ablating ϕ collapses the $k = 0$ model from $R^2 = 0.578$ to -2.79 , confirming that the circular coordinate is doing the work.

2.7 Open issue: the learned coordinate is not yet a gait phase

A per-clip diagnostic of the learned ϕ — revolutions per stride (a true phase completes exactly one), the fraction of steps in which ϕ advances, and the circular correlation with the heuristic phase after removing the constant offset — shows that only steady running approaches a

Table 1: Assist-target R^2 , leave-clips-out. The gap between the heuristic and the learned phase is produced at *equal* information (one scalar).

Model	R^2
Heuristic gait phase	0.096
$S^1 \times \mathbb{R}^0$ (learned phase only)	0.578
$S^1 \times \mathbb{R}^1$	0.613
$S^1 \times \mathbb{R}^2 / \mathbb{R}^3 / \mathbb{R}^4$	0.613 / 0.611 / 0.600
Plain autoencoder, width 1 (no S^1)	0.599
Unconstrained end-to-end	0.629

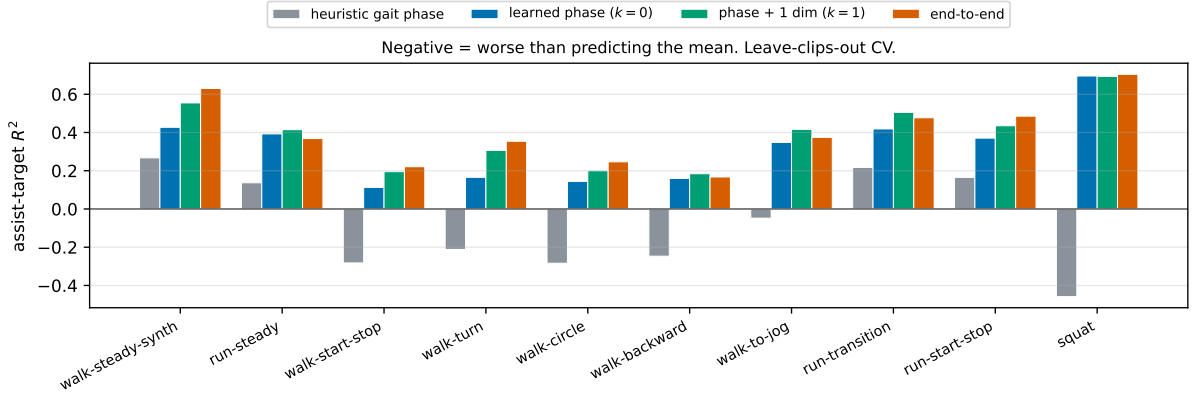


Figure 3: Per-subcategory R^2 . The heuristic phase is *negative* — worse than the mean — in eight of twelve subcategories, while a learned one-dimensional coordinate remains useful throughout.

properly circulating phase (1.11 revolutions per stride, 0.88 monotone, aligned correlation 0.89). Pooled over all clips the median is 0.014 revolutions per stride and 0.539 monotone, i.e. indistinguishable from chance. The monotonicity weight λ_{mono} is evidently too small. At present the claim that survives is therefore the weaker one: a *useful one-dimensional circular code* is learned, not a gait phase. This also explains why $k=0$ attains $R^2 = 0.695$ on squatting, where no gait cycle exists — the coordinate is encoding something other than position within a cycle. Strengthening the constraint and re-examining $k=0$ is the decisive next step.

2.8 Limitations

Running turns (252 samples) and backward running (788) comprise one to two clips each and are negative for every model under leave-clips-out evaluation; they support no conclusion. Stair, slope and jump experts do not yet exist, so the aperiodic end of the spectrum is represented only by squatting, and squatting clips terminate during the first descent in 3 of 10 cases. All results are in simulation with a frozen human policy.

3 Discussion

Interpret the results, describe limitations, and identify next steps.

References

- [1] Leslie Lamport. *LaTeX: A Document Preparation System*. Addison-Wesley, 2 edition, 1994.